

Sai Sri Harsha Guddati

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EDUCATION

University of Florida, Gainesville, FL

Jan 2024 - Dec 2025

Master of Science, Computer Science | CGPA: 3.85/4.0

Courses: Distributed Systems, Computer Networks, Natural Language Processing Applications, Operating Systems, Data Science

Jawaharlal Nehru Technological University, Kakinada, India

Aug 2018 - Jun 2022

Bachelor of Technology, Computer Science | CGPA: 3.8/4.0

Courses: Data Structures & Algorithms, Database Management Systems, Software Engineering, Artificial Intelligence, Data Engineering

WORK EXPERIENCE

University of Florida | Software Development Intern (May 2025 - Present)

Gainesville, FL

- Architected and delivered end-to-end full-stack features across **React and FastAPI/Flask**, owning system design, API contracts, schema evolution, and production rollout for applications serving **30K+ users**.
- Designed a zero-trust authentication and **RBAC system** using **JWT, OAuth2, and role-scoped policies**, eliminating privilege escalation paths and reducing security incidents by 40%.
- Reduced **P95 API latency by 45%** by profiling request lifecycles, optimizing thread pools, rewriting N+1 query patterns, and introducing multi-layer caching strategies.
- Refactored synchronous workflows into asynchronous event-driven pipelines using message queues and background workers, **increasing throughput under peak load by 3x** without infrastructure scaling.
- Led migration from VM-based deployments to **containerized microservices on Kubernetes**, implementing rolling updates and health probes to achieve zero-downtime releases.

TCS | Systems Engineer (Aug 2022 - Dec 2023)

Hyderabad, India

- Developed and operated production backend services using Java Spring Boot and C#/.NET supporting enterprise platforms with **100K+ daily interactions** across business units.
- Designed **backward-compatible REST APIs** with strict validation and idempotent request handling, enabling parallel frontend and integration development without breaking changes.
- Reduced average API response times by **50% by tuning** PostgreSQL and MSSQL query execution plans, optimizing indexes, and eliminating inefficient data access patterns.
- Standardized CI/CD pipelines using GitLab CI, automating builds, tests, and multi-environment deployments, supporting weekly production releases with **zero Sev-1 incidents over 12 months**.
- Converted manual deployment processes into repeatable pipelines, **reducing rollback events by 70%** and significantly improving release reliability.
- Built and optimized background processing workflows to **handle peak data ingestion** without degrading user-facing performance.

PROJECTS

Distributed Metrics Aggregation Engine

Go, PostgreSQL, Redis, Docker

- Designed and implemented a high-throughput metrics ingestion service in **Go using goroutines and channels** for concurrent event processing, time-series aggregation, and indexed query optimization, **reducing dashboard query latency by 60%** compared to naive aggregation approaches.
- Built a stateless REST API layer with connection pooling and Redis-based caching, eliminating redundant database scans and **improving response times by 35–45%** under burst traffic conditions while sustaining stable performance under concurrent load.

ProTerminal — Real-Time Crypto Trading Platform

Python, FastAPI, React, Kafka, Redis, Docker

- Designed and implemented a distributed streaming system ingesting live Binance trade data at sustained high event rates, processing thousands of trades per minute with end-to-end update latency **consistently under 200ms**.
- Architected an event-driven pipeline using Kafka for durable ingestion and Redis Pub/Sub for real-time fan-out, handling burst **traffic spikes of 3–4x without message loss** or consumer lag.
- Optimized React rendering and state management to handle continuous streaming updates, reducing unnecessary re-renders by **~40% and maintaining sub-100ms** UI refresh times under sustained load.

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, Go, JavaScript, Rust, C#, Scala, TypeScript, HTML, CSS, SQL, PHP, Kotlin, R

Frameworks & Libraries: Angular, React, NextJS, Spring Boot, Node.js, Vue, JVM, Redux, Flask, PyTorch

Cloud & Distributed Systems: AWS (ECS, Lambda, DynamoDB, SQS), Azure, GCP, Kubernetes, Docker, Redis, Spark, CI/CD

Databases: PostgreSQL, MySQL, NoSQL, Oracle, MongoDB, Cassandra, Hadoop, BigData

Tools & Platforms: Git/GitHub, GitOps, Selenium, Postman, GraphQL, Jira, Splunk, Tableau, Unix/ Linux, UI/UX